Semester	B.E. Semester VII
Subject	R Programming Lab
Subject Professor In-charge	Prof. Prakash Parmar
Laboratory	M411A

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Experiment No.	01
Experiment Title	Air quality data cleaning using R programming

Task 1 –

RGui - [Untitled - R Editor]

File Edit Packages Windows Help

```
data <- tryCatch({

  read.csv(file.choose(), stringsAsFactors = FALSE)

}, error = function(e) {

  message("Error: ", e$message)
  return(NULL)

})
if (!is.null(data)) {

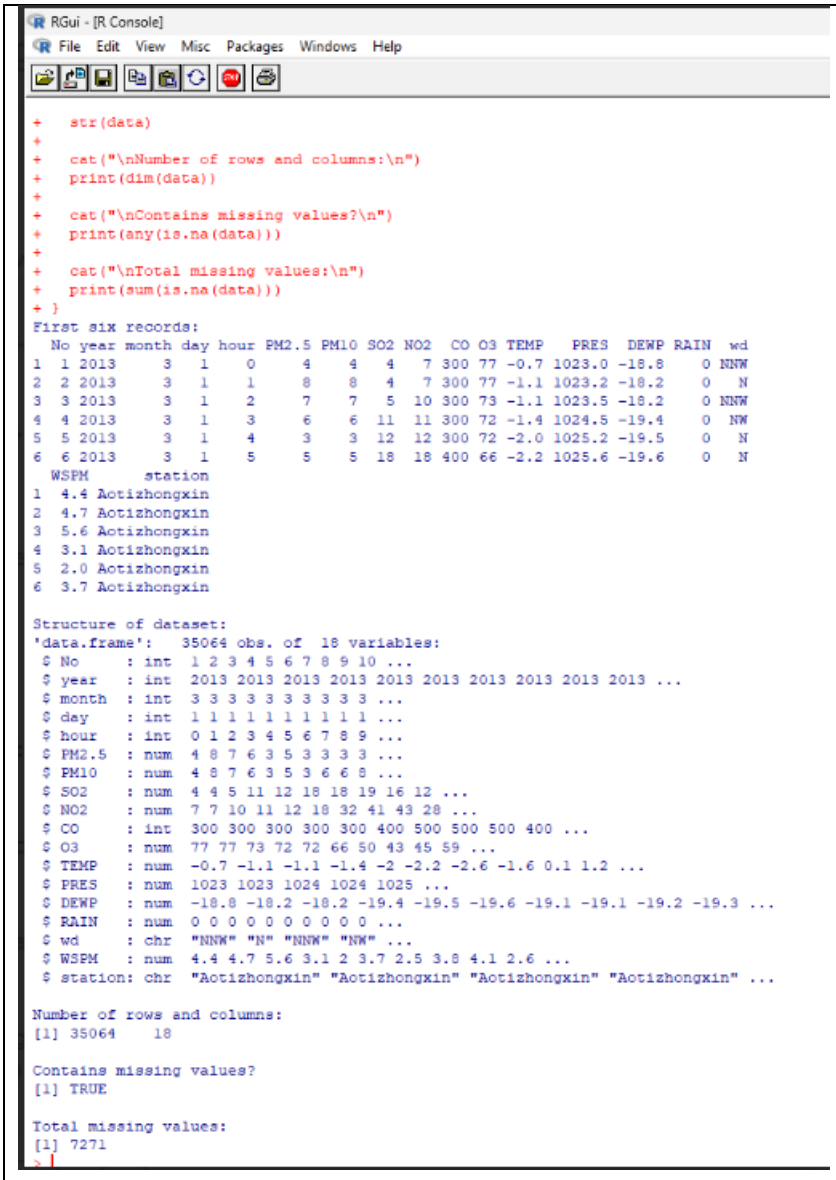
  cat("First six records:\n")
  print(head(data))

  cat("\nStructure of dataset:\n")
  str(data)

  cat("\nNumber of rows and columns:\n")
  print(dim(data))

  cat("\nContains missing values?\n")
  print(any(is.na(data)))

  cat("\nTotal missing values:\n")
  print(sum(is.na(data)))
}
```



Task 2 –

```
#Task 2
#Checking for NA

temperature <- c(28, 30, NA, 32)

cat("Temperature vector:\n")
print(temperature)

cat("\nChecking for NA:\n")
print(is.na(temperature))
```

```
> temperature <- c(28, 30, NA, 32)
>
> cat("Temperature vector:\n")
Temperature vector:
> print(temperature)
[1] 28 30 NA 32
>
> cat("\nChecking for NA:\n")

Checking for NA:
> print(is.na(temperature))
[1] FALSE FALSE  TRUE FALSE
>
```

```r
#Cheacking for NULL

missing_object <- NULL

cat("\nValue of missing_object:\n")
print(missing_object)

cat("\nChecking for NULL:\n")
print(is.null(missing_object))
```

```
> missing_object <- NULL
>
> cat("\nValue of missing_object:\n")

Value of missing_object:
> print(missing_object)
NULL
>
> cat("\nChecking for NULL:\n")

Checking for NULL:
> print(is.null(missing_object))
[1] TRUE
>
```

```r
#CHecking for NaN

undefined_value <- 0 / 0

cat("\nValue of undefined_value:\n")
print(undefined_value)

cat("\nChecking for NaN:\n")
print(is.nan(undefined_value))
```

```
> undefined_value <- 0 / 0
>
> cat("\nValue of undefined_value:\n")

Value of undefined_value:
> print(undefined_value)
[1] NaN
>
> cat("\nChecking for NaN:\n")

Checking for NaN:
> print(is.nan(undefined_value))
[1] TRUE
>
```

Task 3 –

```
#Task 3

missing_summary <- function(df) {

  # Variables to analyze
  vars <- c("PM2.5", "PM10", "SO2", "NO2", "TEMP", "WSPM", "wd")

  # Create an empty data frame for results
  summary_df <- data.frame(
    Variable = character(),
    Total_Records = numeric(),
    Missing_Values = numeric(),
    Missing_Percentage = numeric(),
    stringsAsFactors = FALSE
  )

  # Loop through each variable
  for (var in vars) {

    # Check if variable exists
    if (!(var %in% names(df))) {
      warning(paste("Variable", var, "not found in dataset."))
      next
    }

    total <- nrow(df)
    missing <- sum(is.na(df[[var]]))
    percent <- (missing / total) * 100

    # Add to summary table
    summary_df <- rbind(summary_df,
                        data.frame(
                          Variable = var,
                          Total_Records = total,
                          Missing_Values = missing,
                          Missing_Percentage = round(percent, 2)
                        ))

    # Warning if missing percentage > 20%
    if (percent > 20) {
      warning(paste("Variable", var, "contains more than 20% missing values."))
    }
  }

  return(summary_df)
}

result <- missing_summary(data)
print(result)
```

```
> print(result)
  Variable Total_Records Missing_Values Missing_Percentage
1    PM2.5         35064            925                2.64
2     PM10         35064            718                2.05
3      SO2         35064            935                2.67
4      NO2         35064           1023                2.92
5     TEMP         35064             20                0.06
6     WSPM         35064             14                0.04
7       wd         35064             81                0.23
>
```

Task 4 –

```
#Task 4

# Create pollution ratio
data$pollution_ratio <- data$PM2.5 / data$PM10

# Check invalid values
cat("NA values:", sum(is.na(data$pollution_ratio)), "\n")
cat("NaN values:", sum(is.nan(data$pollution_ratio)), "\n")
cat("Infinite values:", sum(is.infinite(data$pollution_ratio)), "\n")
cat("Positive Infinity:", sum(data$pollution_ratio == Inf, na.rm = TRUE), "\n")
cat("Negative Infinity:", sum(data$pollution_ratio == -Inf, na.rm = TRUE), "\n")

# Replace NaN and Infinite with NA
data$pollution_ratio[is.nan(data$pollution_ratio)] <- NA
data$pollution_ratio[is.infinite(data$pollution_ratio)] <- NA

# Verify
cat("\nAfter replacement:\n")
cat("NaN values:", sum(is.nan(data$pollution_ratio)), "\n")
cat("Infinite values:", sum(is.infinite(data$pollution_ratio)), "\n")
cat("Total NA values:", sum(is.na(data$pollution_ratio)), "\n")
```

```
> # Create pollution ratio
> data$pollution_ratio <- data$PM2.5 / data$PM10
>
> # Check invalid values
> cat("NA values:", sum(is.na(data$pollution_ratio)), "\n")
NA values: 934
> cat("NaN values:", sum(is.nan(data$pollution_ratio)), "\n")
NaN values: 0
> cat("Infinite values:", sum(is.infinite(data$pollution_ratio)), "\n")
Infinite values: 0
> cat("Positive Infinity:", sum(data$pollution_ratio == Inf, na.rm = TRUE), "\n")
Positive Infinity: 0
> cat("Negative Infinity:", sum(data$pollution_ratio == -Inf, na.rm = TRUE), "\n")
Negative Infinity: 0
>
> # Replace NaN and Infinite with NA
> data$pollution_ratio[is.nan(data$pollution_ratio)] <- NA
> data$pollution_ratio[is.infinite(data$pollution_ratio)] <- NA
>
> # Verify
> cat("\nAfter replacement:\n")

After replacement:
> cat("NaN values:", sum(is.nan(data$pollution_ratio)), "\n")
NaN values: 0
> cat("Infinite values:", sum(is.infinite(data$pollution_ratio)), "\n")
Infinite values: 0
> cat("Total NA values:", sum(is.na(data$pollution_ratio)), "\n")
Total NA values: 934
>
```

Task 5 –

```r
#Task 5

# Selected numerical variables
numeric_variables <- c("PM2.5", "PM10", "SO2", "NO2", "TEMP", "WSPM")

# Loop through each variable
for (var in numeric_variables) {

  # Check if the column exists
  if (var %in% names(data)) {

    # Count missing values before replacement
    missing_before <- sum(is.na(data[[var]]))

    # Calculate the median (ignoring missing values)
    median_value <- median(data[[var]], na.rm = TRUE)

    # Replace missing values with the median
    data[[var]][is.na(data[[var]])] <- median_value

    # Count missing values after replacement
    missing_after <- sum(is.na(data[[var]]))

    # Display results
    cat("\n")
    cat("Variable:", var, "\n")
    cat("Missing values before:", missing_before, "\n")
    cat("Median used:", median_value, "\n")
    cat("Missing values after:", missing_after, "\n")

  } else {

    # Display message if the column doesn't exist
    cat("\nColumn", var, "does not exist in the dataset.\n")

  }
}
```

```
Variable: PM2.5
Missing values before: 925
Median used: 58
Missing values after: 0

Variable: PM10
Missing values before: 718
Median used: 87
Missing values after: 0

Variable: SO2
Missing values before: 935
Median used: 9
Missing values after: 0

Variable: NO2
Missing values before: 1023
Median used: 53
Missing values after: 0

Variable: TEMP
Missing values before: 20
Median used: 14.5
Missing values after: 0

Variable: WSPM
Missing values before: 14
Median used: 1.4
Missing values after: 0
> 
```

Task 6 –

```
#Task 6

# Function to calculate the mode
calculate_mode <- function(x) {

  # Remove missing values
  x <- x[!is.na(x)]

  # Calculate the mode
  mode_value <- names(sort(table(x), decreasing = TRUE))[1]

  return(mode_value)
}

# Missing values before replacement
missing_before <- sum(is.na(data$wd))

cat("Missing values before replacement:", missing_before, "\n")

# Calculate mode
mode_wd <- calculate_mode(data$wd)

cat("Mode of wd:", mode_wd, "\n")

# Replace missing values with the mode
data$wd[is.na(data$wd)] <- mode_wd

# Missing values after replacement
missing_after <- sum(is.na(data$wd))

cat("Missing values after replacement:", missing_after, "\n")
```

```
> # Function to calculate the mode
> calculate_mode <- function(x) {
+
+   # Remove missing values
+   x <- x[!is.na(x)]
+
+   # Calculate the mode
+   mode_value <- names(sort(table(x), decreasing = TRUE))[1]
+
+   return(mode_value)
+ }
>
> # Missing values before replacement
> missing_before <- sum(is.na(data$wd))
>
> cat("Missing values before replacement:", missing_before, "\n")
Missing values before replacement: 81
>
> # Calculate mode
> mode_wd <- calculate_mode(data$wd)
>
> cat("Mode of wd:", mode_wd, "\n")
Mode of wd: NE
>
> # Replace missing values with the mode
> data$wd[is.na(data$wd)] <- mode_wd
>
> # Missing values after replacement
> missing_after <- sum(is.na(data$wd))
>
> cat("Missing values after replacement:", missing_after, "\n")
Missing values after replacement: 0
>
```

Task 7 –

```r
#Task 7

clean_variable <- function(df, var_name) {

  tryCatch({

    # Check if the variable exists
    if (!(var_name %in% names(df))) {
      stop("Variable does not exist.")
    }

    # Check if the variable is numeric
    if (!is.numeric(df[[var_name]])) {
      stop("Variable is not numerical.")
    }

    # Check if all values are missing
    if (all(is.na(df[[var_name]]))) {
      stop("Variable contains only missing values.")
    }

    # Calculate the median
    median_value <- median(df[[var_name]], na.rm = TRUE)

    # Check if the median is valid
    if (is.na(median_value) || is.nan(median_value)) {
      stop("Median cannot be calculated.")
    }

    # Replace missing values with the median
    df[[var_name]][is.na(df[[var_name]])] <- median_value

    cat("Variable", var_name, "cleaned successfully.\n")

    # Return the cleaned variable
    return(df[[var_name]])

  }, error = function(e) {

    cat("Error:", e$message, "\n")
    return(NULL)

  })
}
```

```
> clean_variable <- function(df, var_name) {
+
+   tryCatch({
+
+     # Check if the variable exists
+     if (!(var_name %in% names(df))) {
+       stop("Variable does not exist.")
+     }
+
+     # Check if the variable is numeric
+     if (!is.numeric(df[[var_name]])) {
+       stop("Variable is not numerical.")
+     }
+
+     # Check if all values are missing
+     if (all(is.na(df[[var_name]]))) {
+       stop("Variable contains only missing values.")
+     }
+
+     # Calculate the median
+     median_value <- median(df[[var_name]], na.rm = TRUE)
+
+     # Check if the median is valid
+     if (is.na(median_value) || is.nan(median_value)) {
+       stop("Median cannot be calculated.")
+     }
+
+     # Replace missing values with the median
+     df[[var_name]][is.na(df[[var_name]])] <- median_value
+
+     cat("Variable", var_name, "cleaned successfully.\n")
+
+     # Return the cleaned variable
+     return(df[[var_name]])
+
+   }, error = function(e) {
+
+     cat("Error:", e$message, "\n")
+     return(NULL)
+
+   })
+ }
>
>
```

Task 8 –

```
#Task 8

# Selected variables
variables <- c("PM2.5", "PM10", "SO2", "NO2", "TEMP", "WSPM", "wd")

# Create empty comparison table
comparison <- data.frame(
  Variable = character(),
  Missing_Before = numeric(),
  Missing_After = numeric(),
  Values_Replaced = numeric(),
  stringsAsFactors = FALSE
)

# Loop through variables
for (var in variables) {

  # Count missing values before cleaning
  missing_before <- sum(is.na(original_data[[var]]))

  # Count missing values after cleaning
  missing_after <- sum(is.na(data[[var]]))

  # Calculate replaced values
  replaced <- missing_before - missing_after

  # Add results
  comparison <- rbind(comparison,
                      data.frame(
                        Variable = var,
                        Missing_Before = missing_before,
                        Missing_After = missing_after,
                        Values_Replaced = replaced
                      ))
}

# Display comparison table
print(comparison)
```

```
> # Selected variables
> variables <- c("PM2.5", "PM10", "SO2", "NO2", "TEMP", "WSPM", "wd")
>
> # Create empty comparison table
> comparison <- data.frame(
+   Variable = character(),
+   Missing_Before = numeric(),
+   Missing_After = numeric(),
+   Values_Replaced = numeric(),
+   stringsAsFactors = FALSE
+ )
>
> # Loop through variables
> for (var in variables) {
+
+   # Count missing values before cleaning
+   missing_before <- sum(is.na(original_data[[var]]))
+
+   # Count missing values after cleaning
+   missing_after <- sum(is.na(data[[var]]))
+
+   # Calculate replaced values
+   replaced <- missing_before - missing_after
+
+   # Add results
+   comparison <- rbind(comparison,
+                       data.frame(
+                         Variable = var,
+                         Missing_Before = missing_before,
+                         Missing_After = missing_after,
+                         Values_Replaced = replaced
+                       ))
+ }
Error: object 'original_data' not found
>
> # Display comparison table
> print(comparison)
[1] Variable        Missing_Before  Missing_After   Values_Replaced
<0 rows> (or 0-length row.names)
>
```

Task 10 –

```
#Task 10

# Export cleaned dataset
write.csv(
  data,
  "cleaned_air_quality_data.csv",
  row.names = FALSE
)

# Verify export
if (file.exists("cleaned_air_quality_data.csv")) {
  cat("Cleaned dataset exported successfully.\n")
} else {
  cat("Export failed.\n")
}
```

```
> # Export cleaned dataset
> write.csv(
+   data,
+   "cleaned_air_quality_data.csv",
+   row.names = FALSE
+ )
>
> # Verify export
> if (file.exists("cleaned_air_quality_data.csv")) {
+   cat("Cleaned dataset exported successfully.\n")
+ } else {
+   cat("Export failed.\n")
+ }
Cleaned dataset exported successfully.
>
```